UNIVERSIDAD DEL VALLE DE GUATEMALA

Colegio Universitario



Lab No. 3.2 - Algoritmos de Enrutamiento

Catedrático: Kevin Velásquez

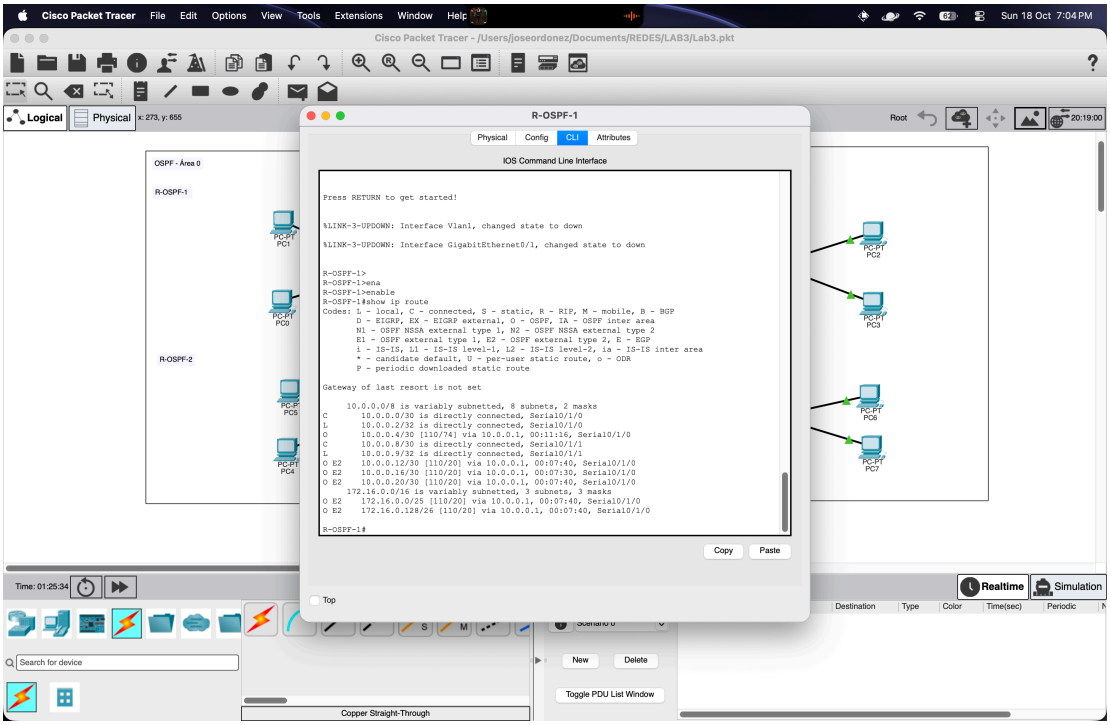
Jose Ordoñez – 231329

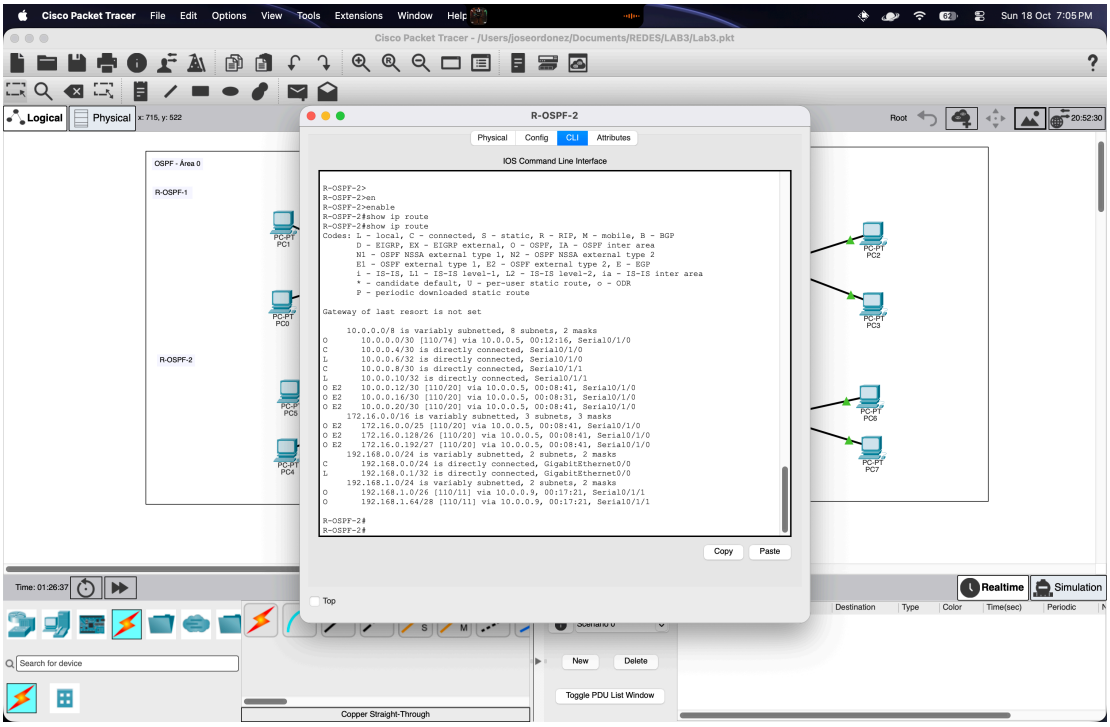
Redes

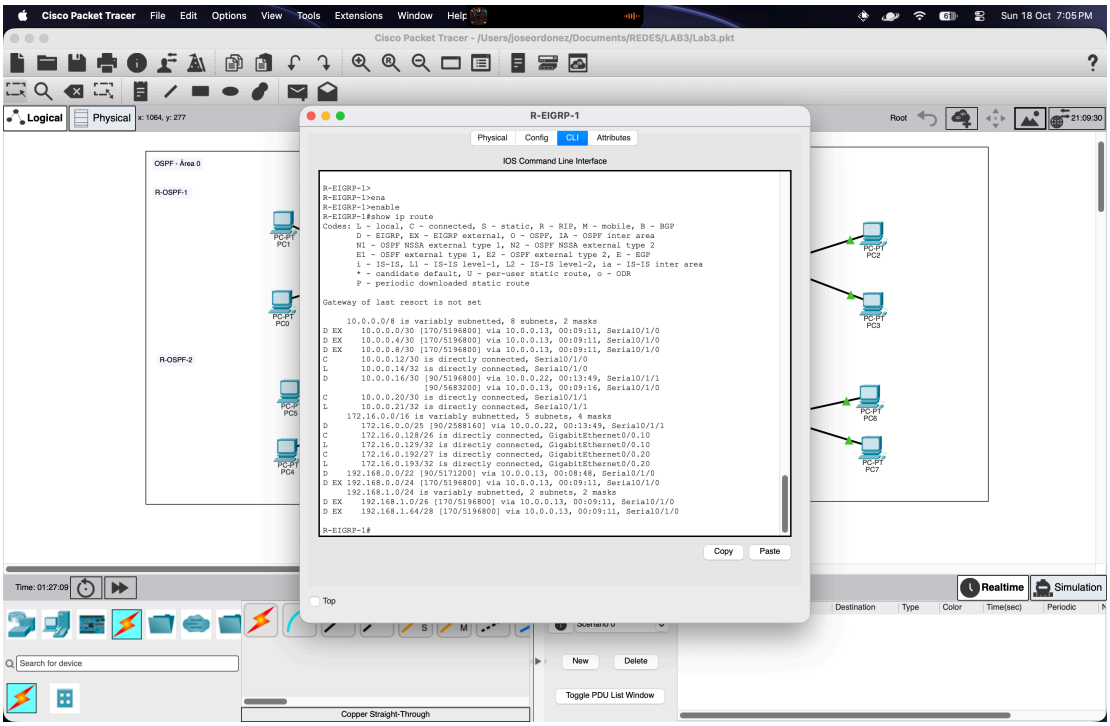
Sección 10

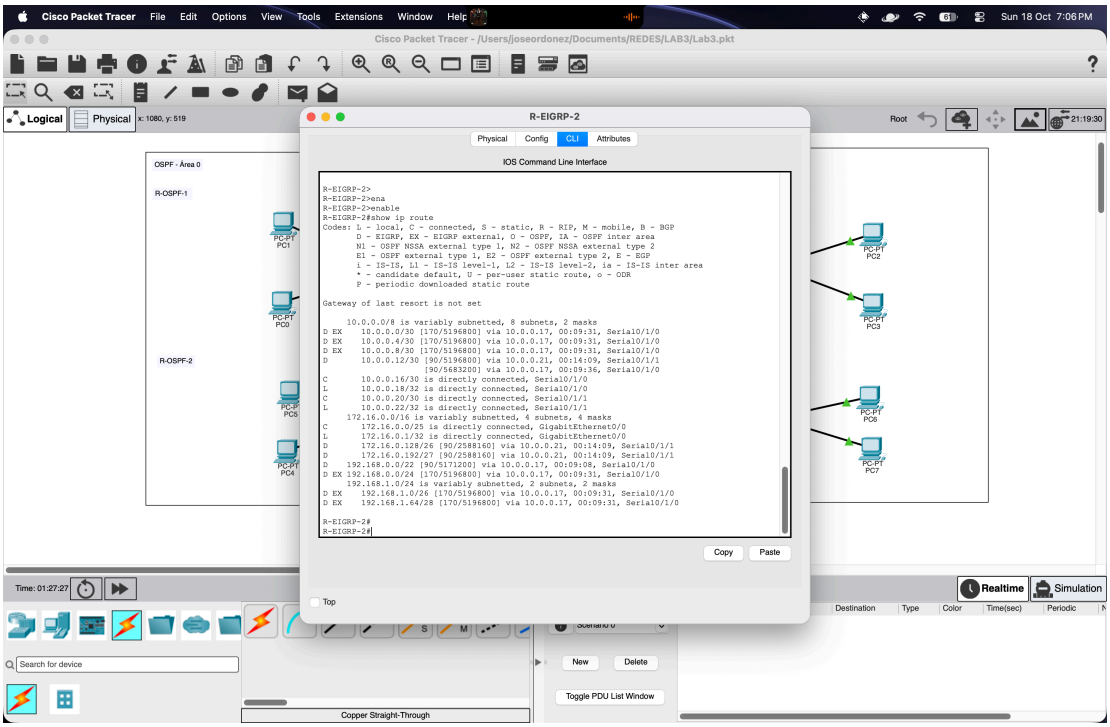
Guatemala, 2026

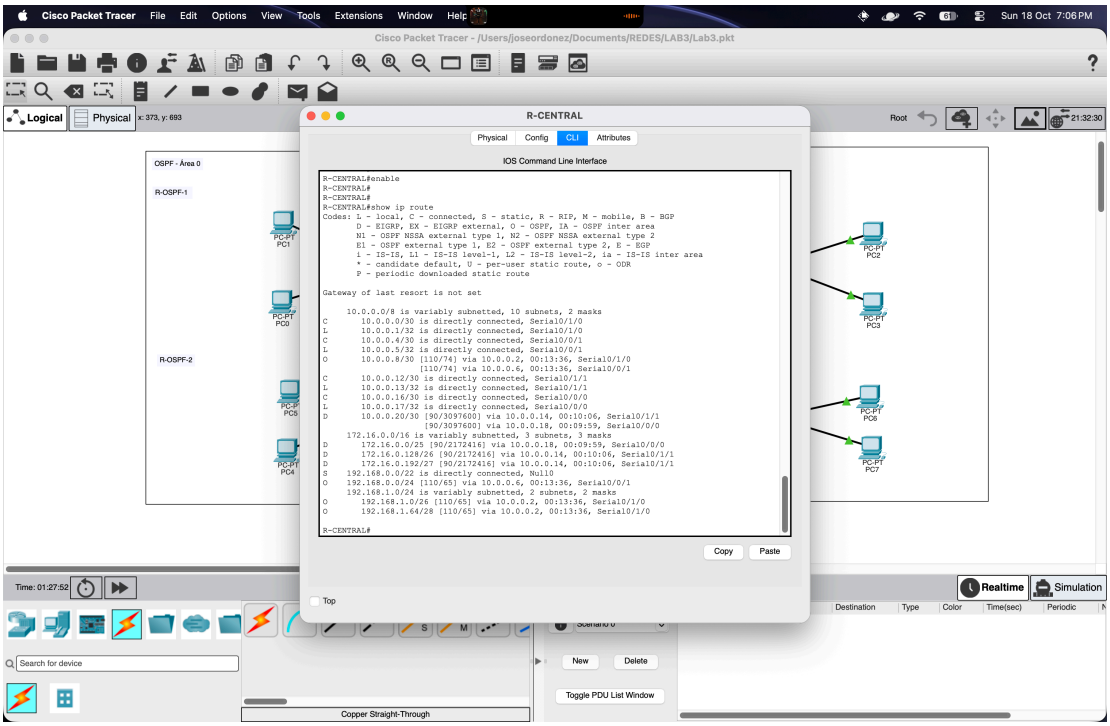
Tablas de enrutamiento:



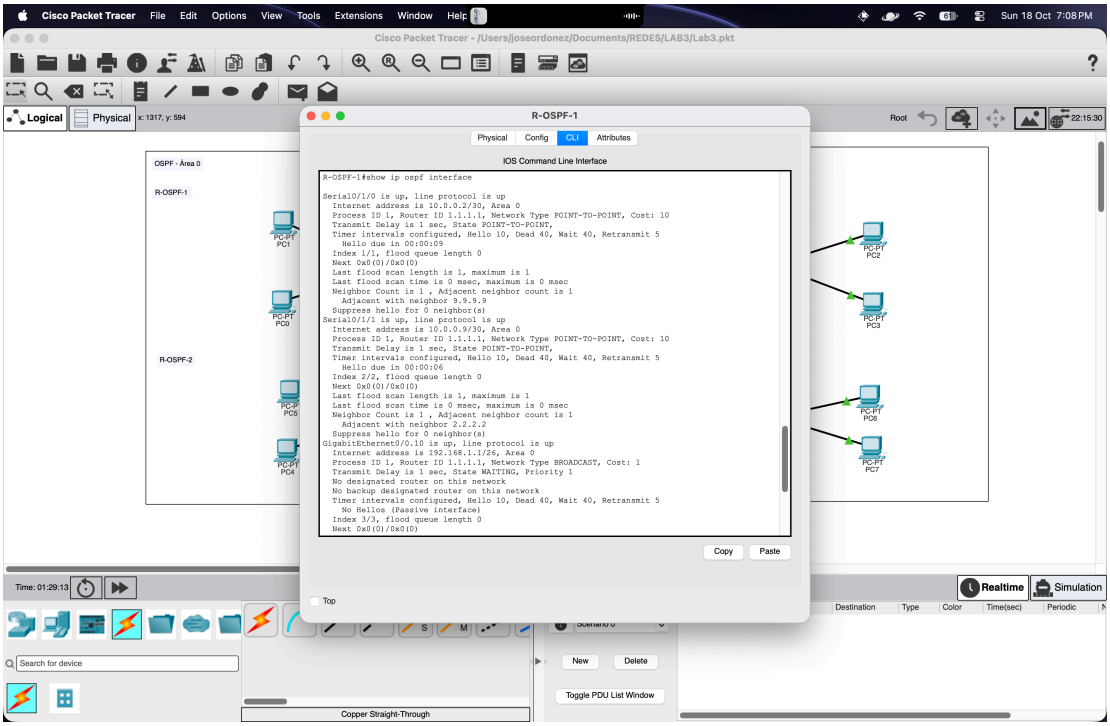



Cisco Packet Tracer File Edit Options View Tools Extensions Window Help Sun 18 Oct 7:05 PM
Cisco Packet Tracer - /Users/joseordonez/Documents/REDES/LAB3/Lab3.pkt
Logical Physical x: 1064, y: 277
Root 21:09:30
R-EIGRP-1
Physical Config CLI Attributes
IOS Command Line Interface
R-EIGRP-1>
R-EIGRP-1>ena
R-EIGRP-1>enable
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX 10.0.0.0/30 [170/5196800] via 10.0.0.13, 00:09:11, Serial0/1/0
D EX 10.0.0.4/30 [170/5196800] via 10.0.0.13, 00:09:11, Serial0/1/0
D EX 10.0.0.8/30 [170/5196800] via 10.0.0.13, 00:09:11, Serial0/1/0
C 10.0.0.12/30 is directly connected, Serial0/1/0
L 10.0.0.14/32 is directly connected, Serial0/1/0
D 10.0.0.16/30 [90/5196800] via 10.0.0.22, 00:13:49, Serial0/1/1
[90/5683200] via 10.0.0.13, 00:09:16, Serial0/1/0
C 10.0.0.20/30 is directly connected, Serial0/1/1
L 10.0.0.21/32 is directly connected, Serial0/1/1
172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
D 172.16.0.0/25 [90/2588160] via 10.0.0.22, 00:13:49, Serial0/1/1
C 172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L 172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C 172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L 172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D 192.168.0.0/22 [90/5171200] via 10.0.0.13, 00:08:48, Serial0/1/0
D EX 192.168.0.0/24 [170/5196800] via 10.0.0.13, 00:09:11, Serial0/1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
D EX 192.168.1.0/26 [170/5196800] via 10.0.0.13, 00:09:11, Serial0/1/0
D EX 192.168.1.64/28 [170/5196800] via 10.0.0.13, 00:09:11, Serial0/1/0
R-EIGRP-1#
Copy Paste
Top
OSPF - Area 0
R-OSPF-1
R-OSPF-2
PC-PT PC1
PC-PT PC0
PC-PT PC5
PC-PT PC4
PC-PT PC2
PC-PT PC3
PC-PT PC6
PC-PT PC7
Time: 01:27:09
Realtime Simulation
Destination Type Color Time(sec) Periodic
Search for device
New Delete
Toggle PDU List Window
Copper Straight-Through

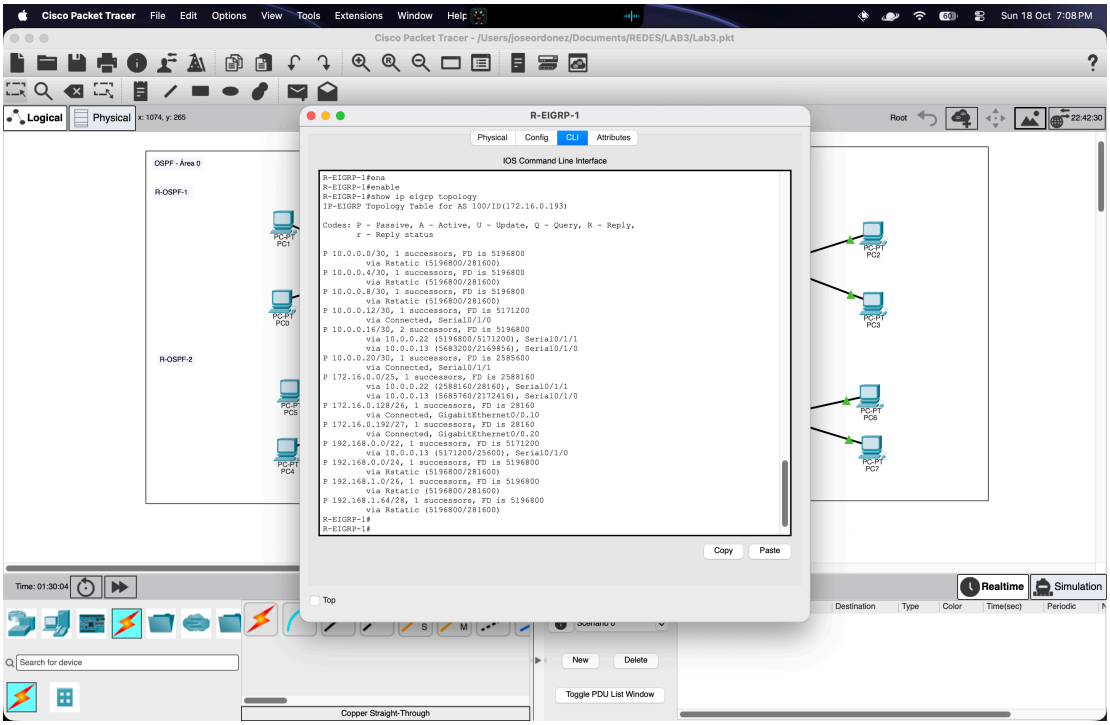

Cisco Packet Tracer File Edit Options View Tools Extensions Window Help Sun 18 Oct 7:06 PM
Cisco Packet Tracer - /Users/joseordonez/Documents/REDES/LAB3/Lab3.pkt
Logical Physical x: 1080, y: 519
Root 21:19:30
R-EIGRP-2
Physical Config CLI Attributes
IOS Command Line Interface
R-EIGRP-2>
R-EIGRP-2>ena
R-EIGRP-2>enable
R-EIGRP-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX 10.0.0.0/30 [170/5196800] via 10.0.0.17, 00:09:31, Serial0/1/0
D EX 10.0.0.4/30 [170/5196800] via 10.0.0.17, 00:09:31, Serial0/1/0
D EX 10.0.0.8/30 [170/5196800] via 10.0.0.17, 00:09:31, Serial0/1/0
D 10.0.0.12/30 [90/5196800] via 10.0.0.21, 00:14:09, Serial0/1/1
[90/5683200] via 10.0.0.17, 00:09:36, Serial0/1/0
C 10.0.0.16/30 is directly connected, Serial0/1/0
L 10.0.0.18/32 is directly connected, Serial0/1/0
C 10.0.0.20/30 is directly connected, Serial0/1/1
L 10.0.0.22/32 is directly connected, Serial0/1/1
172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
C 172.16.0.0/25 is directly connected, GigabitEthernet0/0
L 172.16.0.1/32 is directly connected, GigabitEthernet0/0
D 172.16.0.128/26 [90/2588160] via 10.0.0.21, 00:14:09, Serial0/1/1
D 172.16.0.192/27 [90/2588160] via 10.0.0.21, 00:14:09, Serial0/1/1
D 192.168.0.0/22 [90/5171200] via 10.0.0.17, 00:09:08, Serial0/1/0
D EX 192.168.0.0/24 [170/5196800] via 10.0.0.17, 00:09:31, Serial0/1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
D EX 192.168.1.0/26 [170/5196800] via 10.0.0.17, 00:09:31, Serial0/1/0
D EX 192.168.1.64/28 [170/5196800] via 10.0.0.17, 00:09:31, Serial0/1/0
R-EIGRP-2#
R-EIGRP-2#
Copy Paste
Top
OSPF - Area 0
R-OSPF-1
R-OSPF-2
PC-PT PC1
PC-PT PC0
PC-PT PC5
PC-PT PC4
PC-PT PC2
PC-PT PC3
PC-PT PC6
PC-PT PC7
Time: 01:27:27
Realtime Simulation
Destination Type Color Time(sec) Periodic
Search for device
New Delete
Toggle PDU List Window
Copper Straight-Through



Evidencia de balanceo OSPF:



Evidencia de balanceo EIGRP:



Prueba de conectividad extremo a extremo:

